HW Labs 4 10.11.26

(could use P_X but not as simple CDF is my goal and simple to work with → stuff derived from it.

4.1) Which of the functions are valid CDFs? Find the densities.

a) $$F(x) = \begin{cases} 1 - e^{-x^2} & ; x \geq 0 \\ 0 & ; x < 0 \end{cases}$$ satisfies 1), 2) and 3), hence F is a valid CDF

1) $x < 0$ ... increasing!

$x \geq 0$: $\frac{d}{dx} F(x) = -e^{-x^2} \cdot (-2x) = 2xe^{-x^2} \geq 0$

increasing!

for $x \geq 0$: $F(x) \geq 0$, hence F is increasing on $\mathbb{R}$.

2) $\lim_{x \to \infty} F(x) = 1$, $\lim_{x \to -\infty} F(x) = 0$ ✓

3) Right continuity: $\forall x \in \mathbb{R}: \lim_{\varepsilon \downarrow 0} F(x+\varepsilon) = F(x)$

Only potential discontinuity: 0

$\lim_{\varepsilon \downarrow 0} F(0+\varepsilon) = \lim_{\varepsilon \downarrow 0} \left(1 - e^{-(0+\varepsilon)^2}\right) = \lim_{\varepsilon \downarrow 0} \left(1 - \underbrace{e^{-\varepsilon^2}}_{\downarrow 1}\right) = \underbrace{0}_{\parallel F(0)}$ ✓

Density $$p(x) = \begin{cases} 2xe^{-x^2} & ; x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

b) $$F(x) = \begin{cases} e^{-\frac{1}{x}} & ; x > 0 \\ 0 & ; x \leq 0 \end{cases}$$ F is a valid CDF by 1), 2) and 3)

1) $x, y \leq 0$: $F(x) = 0 \leq 0 = F(y)$

$x \leq 0, y > 0$: $0 \leq e^{-\frac{1}{y}}$

$x, y > 0$: $e^{-\frac{1}{x}}$ is an increasing function!

$\frac{d}{dx} e^{-x^{-1}} = e^{-x^{-1}} \cdot x^{-2} \geq 0$!

2) $\lim_{x \to -\infty} F(x) = 0$, $\lim_{x \to \infty} F(x) = \lim_{x \to \infty} e^{-\left(\frac{1}{x}\right) \to 0} = \underline{\underline{1}}$

3) Potential discontinuity in 0!

$$\lim_{\varepsilon \downarrow 0} e^{-\left(\frac{1}{\varepsilon}\right) \to \infty} = 0 = F(0)$$

$$P(x) = \begin{cases} x^{-2} e^{-x} & ; x \geq 0 \\ 0 & ; x < 0 \end{cases}$$

c) $$F(x) = \begin{cases} 0 & ; x \leq 0 \\ 1/3 & ; 0 < x \leq \frac{1}{2} \\ 1 & ; x > \frac{1}{2} \end{cases}$$

$$\lim_{\varepsilon \downarrow 0} F(0+\varepsilon) = \frac{1}{3} \neq F(0) = 0$$

Thus F is not right cont. in every x ⇓ not a CDF

(Just look for the steps!

4.2) X RV, $F_X(x) = \begin{cases} 0 & ; x < 0 \\ \frac{x^2}{2} & ; 0 \leq x < 1 \\ \frac{1}{2} + \frac{p}{2} & ; 1 \leq x < 2 \\ \frac{1}{2} + \frac{p}{2} + \frac{1-p}{2} & ; x \geq 2 \end{cases}$

a) ✓

Mixed RV → combination of discrete and cont. RV.

($\neq$ cont. $\Rightarrow$ F continuous)

Thus X is a mixed RV.

Thus F is not continuous.

X also isn't discrete, as there don't exist $B \in \mathcal{B}_{\mathbb{R}}$ countable s.t. $P_X(B) = 1$.

b) X is mixed RV. Have to find PDF on continuous parts and PMF on discrete parts

continuous on $(-\infty, 1) \cup (1,2) \cup (2, \infty)$

PDF: $$P_X(x) = \begin{cases} 0 & ; x \in (-\infty, 0) \cup (1,2) \cup (2, \infty) \\ x & ; x \in [0,1) \end{cases}$$

PMF / just measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$

Similar to Bernulli

$$\boxed{P_X(x) = \left(\frac{p}{2}\right)^{2-x} \left(\frac{1-p}{2}\right)^{x-1}}$$

$$P_X(\{1\}) = F_X(1) - F_X(1-) = \frac{1}{2} + \frac{p}{2} - \frac{1}{2} = \frac{p}{2}$$

$$P_X(\{2\}) = F_X(2) - F_X(2-) = \frac{1}{2} + \frac{p}{2} + \frac{1-p}{2} - \left(\frac{1}{2} + \frac{p}{2}\right) = \frac{1-p}{2}$$

4.4) X_k, $k = 1, \dots, n$ $\quad X_k \sim \text{Bernoulli}(p)$

$$P_X(x) = p^x (1-p)^{1-x}$$

$$X = \sum_{k=1}^{n} X_k$$

a) CDF of X_k? 2 outcomes - 0 and 1

$$F_X(x) = \begin{cases} 0 ; & x < 0 \\ 1-p ; & 0 \le x < 1 \\ 1 ; & x \ge 1 \end{cases}$$

b) PMF of X? Binomial RV with support on $\{0, 1, 2, \dots, n\}$, parameters $p \in (0,1)$, $n \in \mathbb{N}_0$.

$$X | n, p \sim \text{Binomial}(n, p)$$

$$P_X(x) = P_X(\{x\}) = \binom{n}{x} p^x (1-p)^{n-x}$$

Probability that sum of Bernoulli RV's is x? x of them are 1 and $n-x$ are 0. This can happen $\binom{n}{x}$ different ways. Trials independent → just multiply rule.

c) CDF of X

$$F_X(x) = \sum_{k=0}^{\min\{\lfloor x \rfloor, n\}} \binom{n}{k} p^k (1-p)^{n-k}$$

$$\| $$

$$\sum_{x_i \in \{0,\dots,n\}} P_X(\{x_i\})$$

$$x_i \le x$$

d) ✓